

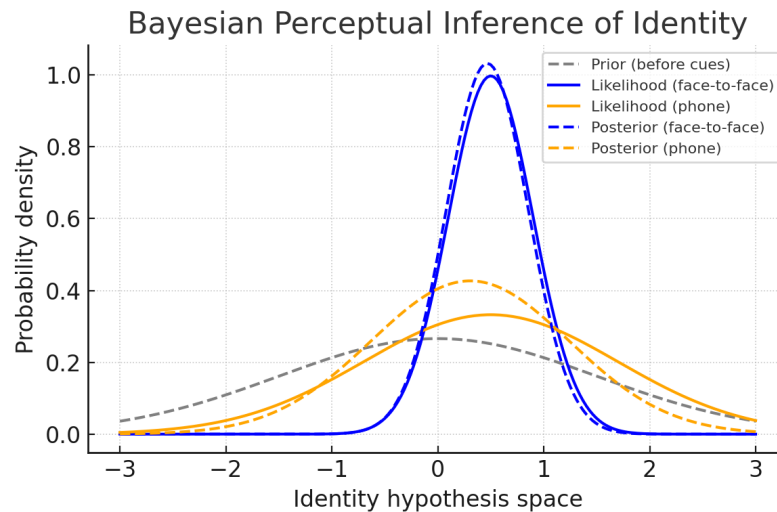
DAEN/ISEN 427 Homework 3

Due date: Tue Sep 23, 2025 (before class)

Rubric: Full credit may be given for any reasonable Bayesian analysis.

1. Why is it that we identify ourselves at the very beginning of a phone conversation, even to people we already know, but we do not do this when we meet in person? Express your answer within the framework of Bayesian perceptual inference.

Answer: In Bayesian perceptual inference, identity recognition combines priors (expectations) with likelihoods (sensory evidence). Face-to-face, the likelihood is strong due to visual and contextual cues, so the posterior probability of correct identification is high, making self-introduction unnecessary. On the phone, the likelihood is weak since only the voice is available, so speakers provide explicit self-identification to strengthen the prior and reduce uncertainty.



2. When a conversation partner speaks softly, or when a conversation occurs in the presence of significant ambient noise, we sometimes cup our ears and/or look carefully at the speaker's lips. Why, in Bayesian perceptual terms, do we do this?

Answer: In Bayesian perceptual terms, understanding speech involves combining prior knowledge (language expectations) with the likelihood provided by sensory evidence. When speech is soft or masked by noise, the auditory likelihood becomes weak and uncertain. By cupping our ears, we increase the auditory signal-to-noise ratio, thereby strengthening the likelihood. By looking at the speaker's lips, we incorporate visual cues, effectively adding another source of likelihood evidence. Both actions reduce uncertainty, allowing a stronger posterior inference about the intended speech.

3. Consider the following simplified version of the broken telephone game. A first player secretly communicates a message $m_1 \in \{0, 1, 2\}$ to a second player with probabilities

$$P(m_1 = 0) = P(m_1 = 1) = P(m_1 = 2) = \frac{1}{3}$$

The second player then secretly communicates a message $m_2 = m_1 + n$ to a third (final) player where

$$P(n = 0) = \frac{2}{3} \quad P(n = +1) = \frac{1}{3}$$

- a. What is the optimal Bayes classification of the message received by the last player?

Answer: Note that $m_2 \in \{0, 1, 2, 3\}$ and

$$P(m_2 = m_1 | m_1) = \frac{2}{3} \quad P(m_2 = m_1 + 1 | m_1) = \frac{1}{3}$$

Hence the posterior distribution is:

$$P(m_1 = 0 | m_2) = \begin{cases} \frac{\frac{2}{3} \cdot \frac{1}{3}}{\frac{2}{3} \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3}} = 1, & m_2 = 0 \\ \frac{\frac{1}{3} \cdot \frac{1}{3}}{\frac{2}{3} \cdot \frac{1}{3} + \frac{1}{3} \cdot \frac{1}{3} + 0 \cdot \frac{1}{3}} = \frac{1}{3}, & m_2 = 1 \\ \frac{0 \cdot \frac{1}{3}}{0 \cdot \frac{1}{3} + \frac{1}{3} \cdot \frac{1}{3} + \frac{2}{3} \cdot \frac{1}{3}} = 0, & m_2 = 2 \\ \frac{0 \cdot \frac{1}{3}}{0 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + \frac{1}{3} \cdot \frac{1}{3}} = 0, & m_2 = 3 \end{cases} \quad P(m_1 = 1 | m_2) = \begin{cases} \frac{0 \cdot \frac{1}{3}}{0 \cdot \frac{1}{3} + \frac{1}{3} \cdot \frac{1}{3} + 0 \cdot \frac{1}{3}} = 0, & m_2 = 0 \\ \frac{\frac{2}{3} \cdot \frac{1}{3}}{\frac{2}{3} \cdot \frac{1}{3} + \frac{1}{3} \cdot \frac{1}{3} + 0 \cdot \frac{1}{3}} = \frac{2}{3}, & m_2 = 1 \\ \frac{\frac{1}{3} \cdot \frac{1}{3}}{\frac{1}{3} \cdot \frac{1}{3} + \frac{2}{3} \cdot \frac{1}{3} + 0 \cdot \frac{1}{3}} = \frac{1}{3}, & m_2 = 2 \\ \frac{0 \cdot \frac{1}{3}}{0 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + \frac{1}{3} \cdot \frac{1}{3}} = 0, & m_2 = 3 \end{cases}$$

and $P(m_1 = 2 | m_2) = 1 - P(m_1 = 0 | m_2) - P(m_1 = 1 | m_2)$

Hence

$$\hat{m}_1(m_2) = \begin{cases} 0, & m_2 = 0 \\ 1, & m_2 = 1 \\ 2, & m_2 = 2 \\ 2, & m_2 = 3 \end{cases}$$

- b. What is the expected error probability?

Answer:

$$P(m_1 = 0 | m_2 = 1)P(m_2 = 1) + P(m_1 = 1 | m_2 = 2)P(m_2 = 2) = \frac{1}{3} \cdot \frac{1}{3} + \frac{1}{3} \cdot \frac{1}{3} = \frac{2}{9}$$